

Spring 26, Ken Short

revised April 19, 2026 7:46 pm

Laboratory 10: Signature Analyzer I - Four Digit Multiplexed Seven-Segment Display Driver

To be performed the week starting April 19th.

Prerequisite Reading

1. Chapter 9 of the textbook.

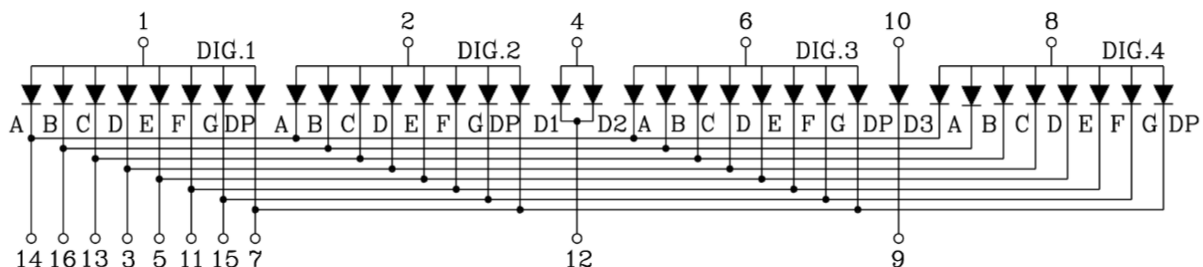
Purpose

The purpose of this laboratory is to design a four digit seven-segment multiplexed LED driver. This system is to be used to display the 16-bit binary input value `signature` as four hexadecimal digits in the funny hex font. The four seven-segment digits of the display are multiplexed. That is, turned ON and then OFF one after another in sequence. When this is done at a high enough rate, all four digits appear to be ON simultaneously.

The LED display that will be used in the test station in the laboratory will be a PARA Light A-394x.



The pinouts for the device are:

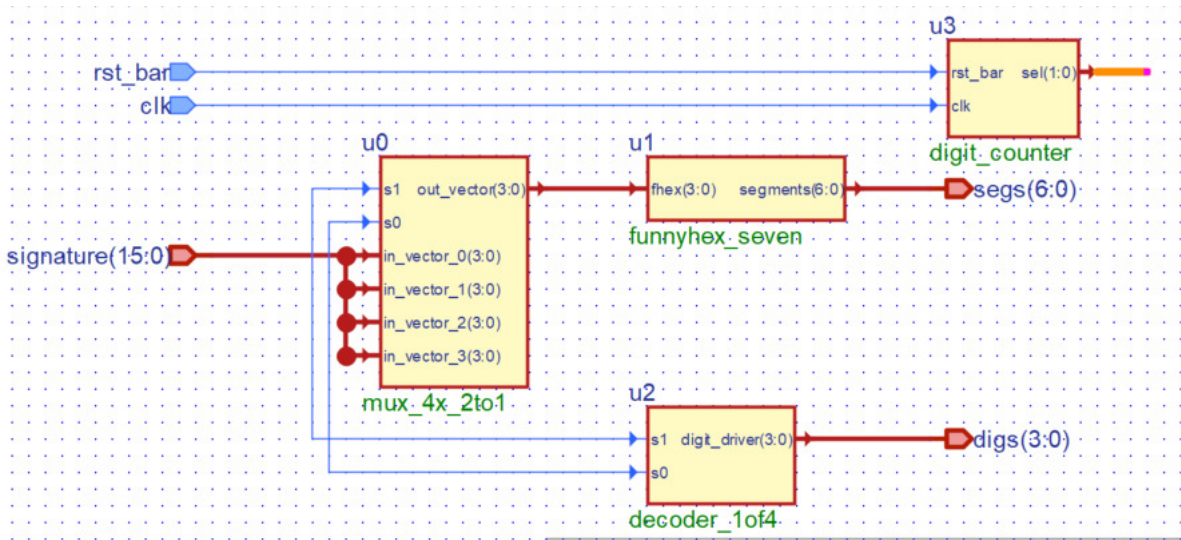


As you can see from the schematic above, six pins (bottom left) are for the segment values. These segment inputs are shared by the four digits. Four pins (1, 2, 6, and 8) are for the digit driver signals that turn individual digits ON. For our work the rightmost digits will be considered digit 0

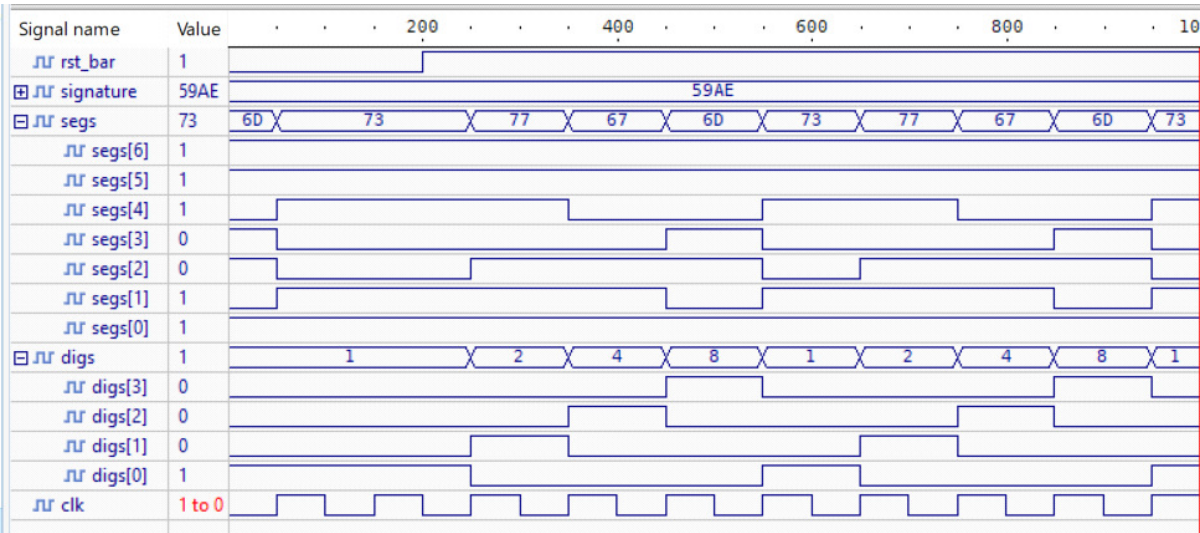
and the leftmost digit will be considered digit 3. The test station will include additional buffers so that a 1 turns ON a digit and a 1 turns ON a segment.

The outputs of the system are `segs (6 downto 0)`, the seven segment values `g` down to `a`, and four signals to turn ON each separate digit, `digs (3 downto 0)`. The 16-bit input data is to be displayed in the funny hex font of Laboratory 05.

A block diagram of the architecture for the system is shown below:



The tasks of this laboratory are to design and verify the four component design entities of the system and the top-level entity that structurally combines them as shown in the previous block diagram. Of the component design entities only `digit_counter` is sequential, the others are combinational.



Tasks

Task 1: *mux_4x_2to1*

Task 2: *funnyhex_seven*

Task 3: *decoder_1of4*

Task 4: *digit_counter*

Task 5: *multiplexed_display_driver*

Design Tasks

You must design the five design entities.

Design Task 1: *mux_4x_2to1*

This design entity is a four to one multiplexer. There are four input channels and one output channel. Each of the channels is 4-bits wide. Each channel carries a hexadecimal value.

```
entity mux_4x_2to1 is
    port(
        s1 : in std_logic;
        s0 : in std_logic;
        in_vector_0 : in std_logic_vector(3 downto 0);
        in_vector_1 : in std_logic_vector(3 downto 0);
        in_vector_2 : in std_logic_vector(3 downto 0);
        in_vector_3 : in std_logic_vector(3 downto 0);
        out_vector : out std_logic_vector(3 downto 0)
    );
end mux_4x_2to1;
```

Write a design description and non-self-checking testbench. Use your testbench to verify your design.

Submit your properly commented design description and testbench source files as part of your prelab.

Design Task 2: *funnyhex_seven*

This design entity is essentially the functionality that you implemented for Laboratory 05, except the inputs and outputs are expressed as vectors.

```
entity funnyhex_seven is
    port(
        fhex : in std_logic_vector(3 downto 0);
        segments : out std_logic_vector(6 downto 0)
    );
end funnyhex_seven;
```

Submit your properly commented design description and testbench source files as part of your prelab.

Design Task 3: *decoder_1of4*

This is a one-out-of- four decoder, also called a two-to-four decoder.

```

entity decoder_lof4 is
    port(
        s1 : in std_logic;
        s0 : in std_logic;
        digit_driver : out std_logic_vector(3 downto 0)
    );
end decoder_lof4;

```

Submit your properly commented design description and testbench source files as part of your prelab.

Design Task 4: digit_counter

This is the only sequential design entity in the system. It is a 2-bit binary counter. It counts from 0 to 3 and then rolls over to 0.

```

entity digit_counter is
    port(
        rst_bar : in std_logic;
        clk : in std_logic;
        sel : out std_logic_vector(1 downto 0)
    );
end digit_counter;

```

Use an unsigned signal to store the internal count. Make the rest signal `rst_bar` synchronous. The counter's `sel` output is used as the select signals to the multiplexer and decoder. Note that on the block diagram these signals are not explicitly shown coming from the counter to the multiplexer and decoder.

Write the design description for `digit_counter`. Use stimulators to verify the counter's operation.

Submit your properly commented design description source file as part of your prelab.

Task 5: multiplexed_display_driver

This is the top-level for the system. It is a structural architecture.

```

entity multiplexed_display_driver is
    port(
        clk : in std_logic;
        rst_bar : in std_logic;
        signature : in std_logic_vector(15 downto 0);
        segs : out std_logic_vector(6 downto 0);
        digs : out std_logic_vector(3 downto 0)
    );
end multiplexed_display_driver;

```

Write a design description for `multiplexed_display_driver`. A testbench will be provided.

Submit your properly commented design description source file as part of your prelab.

Laboratory Tasks

Laboratory Task 1: mux_4x_2to1

Import your VHDL design description files for this design. Use your **testbench** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Laboratory Task 2: funnyhex_seven

Import your VHDL design description files for this design. Use your **testbench** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Laboratory Task 3: decoder_1of4

Import your VHDL design description files for this design. Use your **testbench** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Laboratory Task 4: digit_counter

Import your VHDL design description files for this design. Use **stimulators** to verify your design.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.

Laboratory Task 5: multiplexed_display_driver

Import your VHDL design description files for this design. Use the **provided testbench** to verify your design.

Synthesize and place and route your hp5004a_SA_w_display design description for an ispMACH4A5-64/32-10JC CPLD target.

Demonstrate your timing simulation to a TA and have the TA verify and sign that your timing simulation was successful.

Program your design into an ispMACH4A5-64/32-10JC CPLD.

Demonstrate your simulation of this design to a TA and have the TA verify and sign that your simulation was successful.